

Peripandemic Utilization of Primary Anterior Cruciate Ligament Reconstruction in a United States–Based Integrated Health Care System, 2017–2023

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Abstract

INTRODUCTION: Prior studies reporting the utilization of anterior cruciate ligament reconstruction (ACLR) was limited to pediatric populations or lacked accounting for the impact of the COVID-19 pandemic. In this study, the authors sought to compare the annual incidence of primary ACLR following the COVID-19 pandemic shutdown to see if utilization has rebounded to pre-pandemic levels.

METHODS: The number of ACLRs performed per year from 2017 through 2023 was identified from the authors' health care system's ACLR interregional registry, and average membership per year was identified from membership records. Incidence rates per 100,000 members were calculated for each year. Poisson regression was used to evaluate year-to-year trends overall and across graft selection, age, and gender. Incidence rate ratios (IRRs) and 95% confidence intervals (CIs) are presented.

RESULTS: In total, 23,082 ACLRs were performed from 2017 to 2023. Pre-pandemic, annual ACLR incidence increased from 47.0 per 100,000 in 2017 to 50.2 in 2019. During the pandemic (2020), this dropped by 33.9% to 33.2 per 100,000 (IRR, 0.66; 95% CI, 0.63–0.70). Post-pandemic, annual incidence increased from 38.6 to 42.3 per 100,000 from 2021 to 2023, respectively (2020 to 2021: IRR, 1.16; 95% CI, 1.10–1.23; 2021 to 2022: IRR, 1.11; 95% CI, 1.06–1.16). However, this was still lower than the pre-pandemic incidence. In both genders, patients in the pediatric age groups had the biggest declines during the shutdown (drop off of 57% in the 10- to 14-year-old group and 40% in the 15- to 19-year-old group).

DISCUSSION: The COVID-19 pandemic shutdown of elective orthopedic procedures led to a sharp drop in the numbers of ACLRs performed, and the United States is still in the recovery period.

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Disclosures

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Introduction

Over the past several decades, the incidence of anterior cruciate ligament reconstructions (ACLR) performed has increased. Incidence increased from 17.6 to 50.9 ACLR per 100,000 patients in New York state from 1990 to 2009.¹ Similarly, Canadian ACLR incidence increased from 40.6 to 51.2 per 100,000 patients from 2002 to 2019.² In Norway, the incidence increased from 18 to 26 per 100,000 patients from 2005 to 2021.³ However, the New York and Norwegian studies focused on pediatric patients, while the Canadian study looked at all patients over 10 years of age.

Comparable trends have previously been reported within Kaiser Permanente, a managed health care system in the United States.⁴ Csintalan et al⁴ reported yearly incidence increased from 14.4 ACLRs per 100,000 patients in 2001 to 19.3 ACLRs per 100,000 patients in 2005. However, that study was based on International Classification of Diseases-9 codes without laterality, and, therefore, primary and revision ACLRs could not be delineated. The Kaiser Permanente ACLR Registry was established in 2005, allowing surveillance of all patients undergoing ACLR within the health care system, including the accurate identification of both primary and revision procedures. Of note, the study by Csintalan et al reported ACLR utilization from nearly 20 years ago, and practice patterns have changed over time. This is of particular interest, given the COVID-19 pandemic, which substantially altered patterns for many elective procedures starting in 2020.^{5,6} In the context of total joint arthroplasty, French et al⁵ identified a substantial backlog of patients requiring arthroplasty procedures, with a persistent lag in return to pre-pandemic levels of procedures performed. This suggests the pandemic resulted in a change in practice patterns that is being borne by the patients.

One would expect a similar observation in ACLR practice patterns. Therefore, the present study sought to determine peripandemic changes in ACLR utilization within a large managed health care system in the United States. Year-to-year trends in incidence rates were reported and evaluated for the overall population as well as according to 1) graft selection and 2) patient age and gender. The authors hypothesized there would be a year-to-year difference in utilization of ACLR within the health care system.

Methods

The Strengthening the Reporting of Observational Studies in Epidemiology⁷ checklist was followed for reporting the study. The authors conducted a population-based cohort study. The population was from Kaiser Permanente, a US-based health care system that includes over 12 million members in 8 geographical regions. Patients in the health care system are representative of the areas included.⁸

Kaiser Permanente's ACLR Registry (ACLRR)⁹⁻¹¹ and membership files were the 2 data sources for the study. The ACLRR is a prospective surveillance tool that tracks a predefined set of data elements for all ACLRs performed in the organization. All ACLRs performed within the health care system are identified using procedure codes. This procedure information, as well as additional details, such as graft selection, is reported to the ACLRR by the operating surgeon using intraoperative forms. This information is supplemented with data from the integrated electronic health record and other institutional data sources. Health care plan membership files were used to identify the population of interest; details on membership according to age and gender have been available since 2015.

STUDY SAMPLE

Patients who were health care plan members from 2017 through 2023 and 10-59 years old comprised the study sample. The Georgia and Washington state regions were excluded because they were not part of the health care system for all years of the present study. Annual membership was based on the average monthly membership. The overall annual health care plan membership, as well as membership categorized by age and gender, is presented in Table 1.

EXPOSURE OF INTEREST

Operative year was evaluated discretely.

OUTCOME OF INTEREST

The outcome of interest was undergoing a primary ACLR. ACLR procedures were identified through the ACLRR; revision procedures were excluded. ACLRs were also categorized by graft selection and included allograft, bone-patellar tendon-bone (BPTB) autograft, hamstring (HT) autograft, hybrid graft (defined as using both autograft and allograft), and quadriceps tendon (QT) autograft.

Characteristic	2017	2018	2019	2020	2021	2022	2023
Total N	72.8	75.5	76.1	76.9	77.1	77.5	77.2
Age group, in y, n per 100,000 (%)							
10–14	6.5 (8.9)	6.6 (8.8)	6.7 (8.8)	6.7 (8.7)	6.7 (8.6)	6.6 (8.5)	6.5 (8.4)
15–19	6.8 (9.3)	6.8 (9.1)	6.8 (9.0)	6.8 (8.9)	6.9 (8.9)	7.0 (9.0)	7.0 (9.0)
20–24	7.4 (10.1)	7.5 (10.0)	7.5 (9.9)	7.5 (9.8)	7.5 (9.7)	7.5 (9.7)	7.4 (9.6)
25–29	7.5 (10.3)	7.9 (10.5)	7.9 (10.4)	8.0 (10.3)	7.9 (10.2)	7.8 (10.0)	7.6 (9.8)
30–34	7.7 (10.5)	8.1 (10.7)	8.3 (10.9)	8.6 (11.1)	8.7 (11.3)	8.9 (11.4)	8.8 (11.4)
35–39	7.6 (10.4)	8.0 (10.6)	8.2 (10.8)	8.4 (11.0)	8.5 (11.0)	8.6 (11.0)	8.6 (11.1)
40–44	7.1 (9.7)	7.4 (9.8)	7.6 (9.9)	7.8 (10.1)	8.0 (10.3)	8.2 (10.5)	8.3 (10.7)
45–49	7.4 (10.2)	7.7 (10.2)	7.6 (10.0)	7.6 (9.9)	7.5 (9.8)	7.6 (9.8)	7.6 (9.8)
50–54	7.4 (10.2)	7.6 (10.1)	7.6 (9.9)	7.7 (10.0)	7.8 (10.1)	7.8 (10.1)	7.8 (10.0)
55–59	7.5 (10.3)	7.8 (10.3)	7.8 (10.3)	7.8 (10.2)	7.8 (10.1)	7.7 (9.9)	7.5 (9.7)
Gender, n per 100,000 (%) ^a							
Women	37.2 (51.1)	38.6 (51.1)	38.9 (51.1)	39.3 (51.1)	39.4 (51.2)	39.6 (51.1)	39.4 (50.8)
Men	35.6 (48.9)	36.9 (48.8)	37.2 (48.8)	37.5 (48.8)	37.6 (48.8)	37.8 (48.8)	37.7 (48.6)
Other	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Unknown	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.1 (0.1)	0.1 (0.1)	0.1 (0.1)

Table 1: Health care plan membership per 100,000 by year (2017–2023)

^aSelf-reported.

STATISTICAL ANALYSIS

The annual ACLR incidence was calculated as the number of primary procedures performed divided by the number of health care plan members; rates are presented per 100,000 members. Incidence rates were also calculated according to age, gender, and graft selection. Trends in annual ACLR utilization were assessed using multivariable Poisson regression.¹² The dependent variable was the sum of ACLR annual procedures, the independent variable was operative year, and an offset was included for the total number of operations annually. Overdispersion was evaluated and passed. The model used the prior year as the reference year for each comparison to understand changes over time. Incidence rate ratios (IRRs) and 95% confidence intervals (CIs) are presented; all tests were 2-sided, and a $P < .05$ was the significance threshold. Analyses were performed using R version 3.3.0.

ETHICS STATEMENT

This study was approved by the health care system's institutional review board (#5691). No external funding was received.

Results

Between 2017 and 2023, 23,082 primary ACLRs were performed by 244 surgeons at 52 health

care centers. The mean age of the patients was 28.3 years, and the majority were men (56.6%). The median time from injury to ACLR was 3.4 months in 2019; this increased to 5.0 months in 2020 and recovered to 3.2 months in 2021. For the overall cohort, BPTB was the most common graft (34.1%), followed by HT (29.7%), allograft (22.5%), and QT (10.5%). Characteristics of patients who underwent an ACLR by operative year are presented in Table 2. Graft selection across operative years is provided in Figure 1. HT autograft was the most frequently selected graft until 2019, when BPTB autograft became the most common.

OVERALL ACLR UTILIZATION

The annual incidence of ACLR is presented in Table 3, with a peak incidence of 50.2 per 100,000 members in 2019 (Figure 2). Incidence increased between 2017 and 2018 (IRR, 1.05; 95% CI, 1.00–1.10). There was a decrease between 2019 and 2020 (IRR, 0.66; 95% CI, 0.63–0.70), which corresponded with the suspension of most elective procedures during the COVID-19 pandemic and was followed by a continual increase from 2020 to 2022 (2020 to 2021: IRR, 1.16; 95% CI, 1.10–1.23; 2021 to 2022: IRR, 1.11; 95% CI, 1.06–1.16), with a subsequent leveling off from 2022 to 2023.

Characteristic	2017	2018	2019	2020	2021	2022	2022
Total N	3422	3710	3820	2556	2979	3329	3266
Age, in y, mean (SD)	27.8 (11.2)	28.3 (11.3)	28.6 (11.4)	29.4 (11.2)	28.0 (11.4)	28.2 (11.4)	28.3 (11.6)
Age group, n (%)							
10-14	165 (4.8)	152 (4.1)	158 (4.1)	68 (2.7)	122 (4.1)	146 (4.4)	129 (3.9)
15-19	931 (27.2)	972 (26.2)	979 (25.6)	586 (22.9)	836 (28.1)	904 (27.2)	949 (29.1)
20-24	505 (14.8)	502 (13.5)	520 (13.6)	350 (13.7)	406 (13.6)	448 (13.5)	388 (11.9)
25-29	488 (14.3)	547 (14.7)	552 (14.5)	416 (16.3)	432 (14.5)	457 (13.7)	416 (12.7)
30-34	400 (11.7)	479 (12.9)	486 (12.7)	321 (12.6)	343 (11.5)	434 (13.0)	429 (13.1)
35-39	331 (9.7)	365 (9.8)	396 (10.4)	286 (11.2)	306 (10.3)	319 (9.6)	321 (9.8)
40-44	258 (7.5)	286 (7.7)	289 (7.6)	236 (9.2)	216 (7.3)	251 (7.5)	249 (7.6)
45-49	189 (5.5)	236 (6.4)	232 (6.1)	139 (5.4)	163 (5.5)	195 (5.9)	190 (5.8)
50-54	100 (2.9)	109 (2.9)	138 (3.6)	109 (4.3)	106 (3.6)	115 (3.5)	137 (4.2)
55-59	55 (1.6)	62 (1.7)	70 (1.8)	45 (1.8)	49 (1.6)	60 (1.8)	58 (1.8)
Gender, n (%)							
Women	1466 (42.8)	1555 (41.9)	1642 (43.0)	1099 (43.0)	1348 (45.3)	1432 (43.0)	1479 (45.3)
Men	1956 (57.2)	2155 (58.1)	2178 (57.0)	1456 (57.0)	1631 (54.7)	1897 (57.0)	1786 (54.7)
Unknown	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.0)	0 (0.0)	0 (0.0)	1 (0.0)
Time from injury to ACLR, in months							
Mean (SD)	14.6 (35.3)	14.2 (32.1)	13.0 (32.1)	16.8 (36.4)	14.3 (33.7)	12.6 (30.7)	12.5 (31.1)
Median (IQR)	3.6 (2.2-8.0)	3.7 (2.1-9.0)	3.4 (2.0-7.6)	5.0 (2.7-10.7)	3.2 (2.0-7.8)	3.6 (2.2-7.2)	3.9 (2.3-7.5)
Missing	951 (27.8)	1453 (39.2)	848 (22.2)	401 (15.7)	426 (14.3)	452 (13.6)	445 (13.6)
Graft selection, n (%)							
Allograft	833 (24.3)	931 (25.1)	931 (24.4)	609 (23.8)	587 (19.7)	627 (18.8)	675 (20.7)
BPTB autograft	1048 (30.6)	1124 (30.3)	1294 (33.9)	843 (33.0)	1071 (36.0)	1255 (37.7)	1241 (38.0)
HT autograft	1295 (37.8)	1310 (35.3)	1196 (31.3)	803 (31.4)	803 (27.0)	791 (23.8)	651 (19.9)
Hybrid graft	129 (3.8)	126 (3.4)	119 (3.1)	98 (3.8)	97 (3.3)	72 (2.2)	57 (1.7)
QT autograft	101 (3.0)	212 (5.7)	275 (7.2)	202 (7.9)	417 (14.0)	578 (17.4)	639 (19.6)
Other	3 (0.1)	1 (0.0)	1 (0.0)	1 (0.0)	1 (0.0)	5 (0.2)	3 (0.1)
Unknown	13 (0.4)	6 (0.2)	4 (0.1)	0 (0.0)	3 (0.1)	1 (0.0)	0 (0.0)

Table 2: Characteristics of 23,082 patients who underwent primary anterior cruciate ligament reconstruction within a US-based health care system by year (2017-2023)

ACLR = anterior cruciate ligament reconstruction; BPTB = bone-patellar tendon-bone; HT = hamstring; IQR = interquartile range; QT = quadriceps tendon; SD = standard deviation.

ACLR UTILIZATION BY GRAFT SELECTION

Annual incidence of ACLR by graft selection is presented in Table 3. No change in year-to-year allograft ACLR utilization was observed except for the period from 2019 to 2020 (IRR, 0.64; 95% CI, 0.58-0.71). Greater variation was observed in BPTB ACLR across operative years with no change from 2017 to 2018, an increase from 2018 to 2019 (IRR, 1.15; 95% CI, 1.06-1.24), a decrease from 2019 to 2020 (IRR, 0.65; 95% CI, 0.59-0.71), and then an increase from 2020 to 2022 (2020 to 2021: IRR, 1.27; 95% CI, 1.16-1.39; 2021 to 2022: IRR, 1.16; 95% CI, 1.07-1.26). HT ACLR utilization

decreased from 2018 to 2020 (2018 to 2019: IRR, 0.91; 95% CI, 0.84 -0.98; 2019 to 2020: IRR, 0.67; 95% CI, 0.61-0.73); a decrease was also observed from 2022 to 2023 (IRR, 0.83; 95% CI, 0.74-0.92). Hybrid graft ACLR utilization decreased across all operative years. QT ACLR incidence increased across all years (IRR range, 1.29-2.06) except for a decrease from 2019 to 2020 (IRR, 0.73; 95% CI, 0.61-0.87) and no change from 2022 to 2023. By 2023, QT ACLR utilization was similar to allograft and HT ACLR (Figure 2).

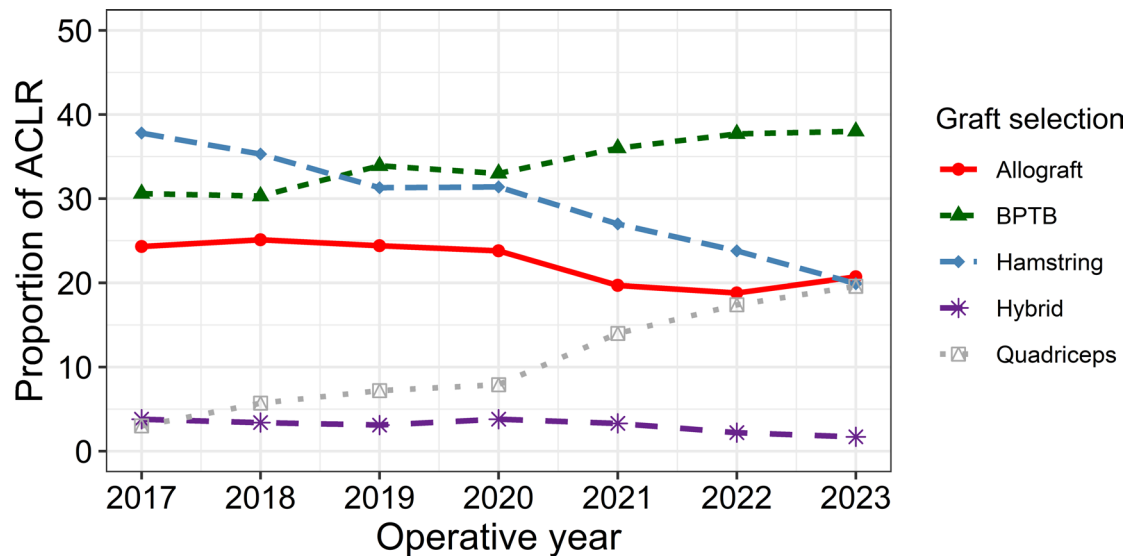


Figure 1: Annual graft selection proportion of anterior cruciate ligament reconstruction across operative years. ACLR = anterior cruciate ligament reconstruction; BPTB = bone-patellar tendon-bone.

AGE- AND GENDER-STRATIFIED ACLR UTILIZATION

The annual incidence of ACLR according to age and gender group is presented in Table 4. Results by gender specifically and age specifically are provided in Supplemental Appendix 1 and 2. Incidence across all operative years was highest for both women and men 15–19 years of age (Figure 3).

The steepest decrease in ACLR utilization for women was observed from 2019 to 2020 (IRR range, 0.36–0.75). The only increase in ACLR incidence was from 2020 to 2021 for women aged 10–14 years (IRR, 2.01; 95% CI, 1.39–2.96) and 15–19 years (IRR, 1.40; 95% CI, 1.21–1.62).

No changes in ACLR utilization were observed for men aged 10–14 and 55–59 years. Any declines in ACLR incidence for other age groups were observed between 2019 and 2020 and between 2022 and 2023. The only increase in ACLR utilization observed for men was for those aged 45–49 years from 2017 to 2018 (IRR, 1.31; 95% CI, 1.00–1.73), aged 15–19 from 2020 to 2022 (2020 to 2021: IRR, 1.44; 95% CI, 1.24–1.68; 2021 to 2022: IRR, 1.16; 95% CI, 1.02–1.33), and aged 30–34 from 2021 to 2022 (IRR, 1.24; 95% CI, 1.04–1.48).

Discussion

The COVID-19 pandemic led to an unprecedented disruption of elective medical procedures.^{5,6,13} The

present study found ACLR incidence declined from 2019 to 2020 during the COVID-19 pandemic by 33.9%. Importantly, by 2023, post-pandemic ACLR utilization had not recovered to pre-pandemic levels for either women or men. A disproportionate decline in ACLR utilization was observed for both genders in the pediatric age groups.

Several pre-pandemic studies reported a trend toward increasing ACLR incidence.^{1,14,15} Mall et al¹⁴ found that the incidence of ACLR rose in the United States from 32.9 to 43.5 per 100,000 person-years from 1994 to 2006. These findings were corroborated by Buller et al¹⁵ In a region-specific study, Dodwell et al¹ searched International Classification of Diseases-9 codes in a New York state database and determined that the incidence of pediatric ACLR rose from 17.6 per 100,000 in 1990 to 50.9 per 100,000 in 2009. The pre-pandemic incidence rates reported in the present study, using data from a health care system predominantly representing the western United States, were in line with these other reports: rising from 47.0 to 50.2 per 100,000 from 2017 to 2019. The same held true for the pediatric population in this study's database, with a rise from 82.4 to 84.2 per 100,000 over the same period.

Surprisingly, there have not been many publications evaluating the impact of COVID-19 on ACLR. Kiani et al⁶ utilized current procedural terminology codes to identify ACLR case volume among pediatric patients in the Pediatric Health Information

Operative year	Incidence per 100,000 members	IRR (95% CI) ^a	P value
<i>Overall</i>			
2017	47.0	—	
2018	49.2	1.05 (1.00–1.10)	.041 ^b
2019	50.2	1.02 (0.98–1.07)	.354
2020	33.2	0.66 (0.63–0.70)	< .001 ^b
2021	38.6	1.16 (1.10–1.23)	< .001 ^b
2022	42.9	1.11 (1.06–1.16)	< .001 ^b
2023	42.3	0.98 (0.94–1.03)	.502
<i>Graft selection</i>			
<i>Allograft</i>			
2017	11.4	—	
2018	12.3	1.07 (0.98–1.18)	.136
2019	12.2	0.99 (0.90–1.08)	.811
2020	7.9	0.64 (0.58–0.71)	< .001 ^b
2021	7.6	0.96 (0.86–1.07)	.458
2022	8.1	1.06 (0.95–1.18)	.325
2023	8.7	1.08 (0.97–1.20)	.178
<i>BPTB autograft</i>			
2017	14.4	—	
2018	14.9	1.04 (0.96–1.14)	.306
2019	17.0	1.15 (1.06–1.24)	.001 ^b
2020	11.0	0.65 (0.59–0.71)	< .001 ^b
2021	13.9	1.27 (1.16–1.39)	< .001 ^b
2022	16.2	1.16 (1.07–1.26)	< .001 ^b
2023	16.1	0.99 (0.92–1.07)	.832
<i>HT autograft</i>			
2017	17.8	—	
2018	17.4	0.98 (0.91–1.06)	.586
2019	15.7	0.91 (0.84–0.98)	.013 ^b
2020	10.4	0.67 (0.61–0.73)	< .001 ^b
2021	10.4	1.00 (0.90–1.10)	.962
2022	10.2	0.98 (0.89–1.08)	.657
2023	8.4	0.83 (0.74–0.92)	< .001 ^b
<i>Hybrid graft^c</i>			
2017	1.8	—	
2018	1.7	0.94 (0.74–1.21)	.638
2019	1.6	0.94 (0.73–1.20)	.608
2020	1.3	0.82 (0.62–1.06)	.134
2021	1.3	0.99 (0.74–1.31)	.927
2022	0.9	0.74 (0.54–1.00)	.051
2023	0.7	0.80 (0.56–1.12)	.197
<i>QT autograft^c</i>			
2017	1.4	—	
2018	2.8	2.03 (1.60–2.58)	< .001 ^b
2019	3.6	1.29 (1.08–1.54)	.006 ^b

Table 3: Crude incidence and adjusted rate ratio of primary anterior cruciate ligament reconstruction within a US-based health care system by year, overall and by graft selection
(Continued)

Table 3: Continued

Operative year	Incidence per 100,000 members	IRR (95% CI) ^a	P value
2020	2.6	0.73 (0.61–0.87)	.001 ^b
2021	5.4	2.06 (1.74–2.44)	< .001 ^b
2022	7.5	1.38 (1.22–1.56)	< .001 ^b
2023	8.3	1.11 (0.99–1.24)	0.067

^aPoisson regression model adjusted for age and gender. The prior year was used as the reference year in regression models.

^bStatistically significant with a $P < .05$, meaning a significant change in incidence from the prior year.

^cCrude regression results presented; too few numbers for age- and gender-adjusted analysis.

BPTB = bone-patellar tendon-bone; CI = confidence interval; HT = hamstring tendon; IRR = incidence rate ratio; QT = quadriceps tendon.

System. During the pandemic, they estimated ACLR volume decreased by 30% from pre-pandemic levels. This is similar to the authors' finding of a 33.9% decline in incidence in their cohort from 2019 to 2020. However, the pediatric patient population in the present study experienced a more pronounced decline from 2019 to 2020 (10- to 14-year-olds: 57%, 15- to 19-year-olds: 40%). The discrepancy in this study's findings can be explained by differing data gathering techniques; the present study utilized granular registry data that made it possible to clearly distinguish between revision and primary procedures. Additionally, Kiani et al⁶ looked at current procedural terminology codes obtained from pediatric hospitals, thereby potentially misrepresenting the true incidence, because high-school-age pediatric patients, particularly women, who are skeletally mature, may have elected to receive surgery at an adult hospital. Nevertheless,

the pediatric population in this study experienced a disproportionate decline in ACLR utilization relative to other age groups, where the next highest decline was seen in patients in their 4th decade of life (21% decline). The disproportionate decline may be explained by the halting of sports participation in addition to the precautions taken by hospital systems to halt elective procedures.^{6,13}

Post-pandemic recovery has been a topic of concern in elective orthopedic procedures.^{5,13,16} French et al⁵ estimate that it could take decades to address the backlog of total joint arthroplasties that resulted from the pandemic shutdown. Delayed surgery in ACLR is also a concern, as a higher risk for further internal derangement of the knee, including chondral and meniscal pathology, has been reported.^{17,18} This study's findings suggest the pandemic had a temporary and likely insignificant

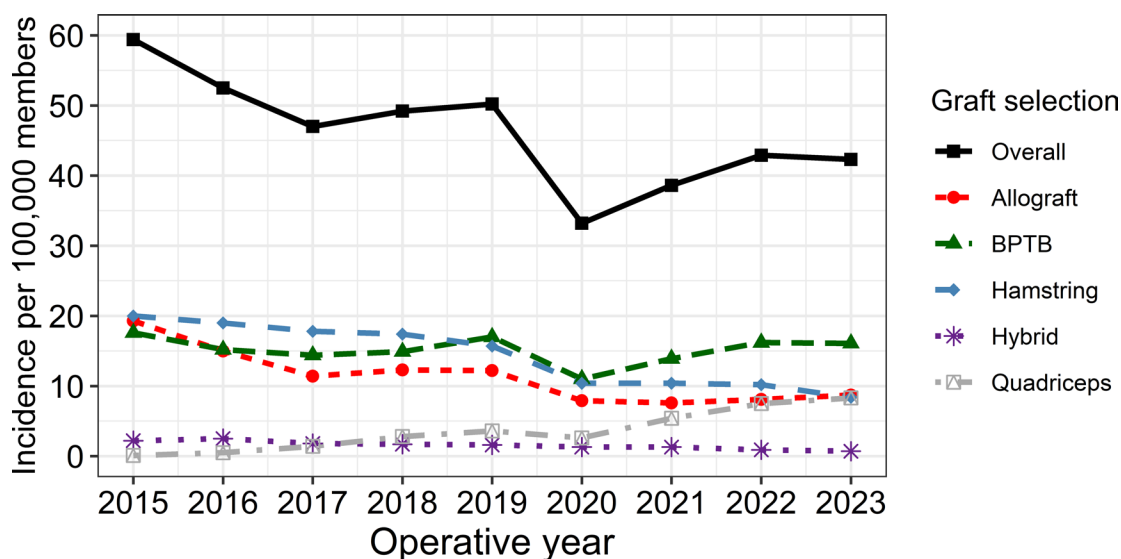


Figure 2: Annual incidence of primary anterior cruciate ligament reconstruction in a US-based health care system, overall and by graft selection (2017–2023). BPTB = bone-patellar tendon-bone.

ACLR Utilization in a US Health Care System

Age groups	Women			Men		
Operative year	Incidence ^a	IRR (95% CI) ^b	P value	Incidence ^a	IRR (95% CI) ^b	P value
10–14 y						
2017	36.1	—		15.4	—	
2018	32.8	0.91 (0.70–1.18)	.468	13.6	0.88 (0.59–1.31)	.529
2019	35.2	1.07 (0.82–1.40)	.599	12.6	0.93 (0.61–1.41)	.720
2020	12.5	0.36 (0.25–0.50)	< .001 ^c	7.9	0.63 (0.38–1.01)	.057
2021	25.2	2.01 (1.39–2.96)	< .001 ^c	11.8	1.49 (0.92–2.45)	.109
2022	27.0	1.07 (0.79–1.45)	.659	17.5	1.49 (1.00–2.24)	.052
2023	26.4	0.98 (0.73–1.32)	.892	13.5	0.77 (0.52–1.14)	.195
15–19 y						
2017	146.7	—		127.8	—	
2018	151.1	1.03 (0.91–1.17)	.646	133.4	1.04 (0.92–1.19)	.521
2019	144.6	0.96 (0.85–1.08)	.491	141.5	1.06 (0.93–1.20)	.360
2020	92.7	0.64 (0.56–0.74)	< .001 ^c	79.1	0.56 (0.48–0.65)	< .001 ^c
2021	130.0	1.40 (1.21–1.62)	< .001 ^c	114.0	1.44 (1.24–1.68)	< .001 ^c
2022	127.7	0.98 (0.86–1.12)	.795	132.4	1.16 (1.02–1.33)	.028 ^c
2023	138.8	1.09 (0.95–1.24)	.213	132.5	1.00 (0.88–1.14)	.992
20–24 y						
2017	40.9	—		96.1	—	
2018	42.3	1.03 (0.83–1.29)	.768	90.9	0.95 (0.82–1.10)	.463
2019	49.9	1.18 (0.96–1.46)	.123	88.5	0.97 (0.84–1.13)	.729
2020	29.9	0.60 (0.47–0.76)	< .001 ^c	63.2	0.71 (0.60–0.84)	< .001 ^c
2021	36.3	1.21 (0.94–1.56)	.135	72.2	1.14 (0.96–1.36)	.135
2022	43.0	1.19 (0.94–1.49)	.142	76.2	1.05 (0.89–1.24)	.532
2023	40.8	0.95 (0.76–1.19)	.644	63.2	0.83 (0.70–0.99)	.033 ^b
25–29 y						
2017	42.3	—		89.9	—	
2018	39.3	0.93 (0.75–1.15)	.500	102.2	1.14 (0.98–1.32)	.088
2019	44.9	1.14 (0.93–1.41)	.217	96.7	0.95 (0.82–1.09)	.446
2020	33.4	0.74 (0.60–0.93)	.008c	73.2	0.76 (0.65–0.88)	< .001c
2021	38.4	1.15 (0.92–1.45)	.222	73.4	1.00 (0.85–1.19)	.974
2022	37.8	0.98 (0.79–1.23)	.874	81.6	1.11 (0.94–1.31)	.202
2023	42.3	1.12 (0.90–1.40)	.312	68.5	0.84 (0.71–0.99)	.040 ^c
30–34 y						
2017	32.4	—		73.5	—	
2018	37.8	1.17 (0.93–1.48)	.188	82.1	1.12 (0.95–1.31)	.178
2019	43.0	1.14 (0.92–1.41)	.232	75.2	0.92 (0.78–1.07)	.269
2020	25.6	0.59 (0.47–0.75)	< .001 ^c	50.4	0.67 (0.56–0.80)	< .001 ^c
2021	27.5	1.08 (0.84–1.39)	.569	52.5	1.04 (0.86–1.26)	.683
2022	34.5	1.25 (0.99–1.59)	.060	64.9	1.24 (1.04–1.48)	.019c
2023	37.0	1.07 (0.86–1.33)	.521	61.6	0.95 (0.80–1.12)	.544
35–39 y						
2017	32.2	—		56.1	—	
2018	33.3	1.03 (0.81–1.32)	.781	58.5	1.04 (0.86–1.26)	.664
2019	33.5	1.01 (0.80–1.27)	.964	63.6	1.09 (0.91–1.30)	.356
2020	29.3	0.87 (0.69–1.11)	.273	39.0	0.61 (0.50–0.75)	< .001 ^b

Table 4: Age- and gender-stratified incidence and rate ratio of primary anterior cruciate ligament reconstruction within a US-based health care system by year (*Continued*)

Table 4: Continued

Age groups	Women			Men		
2021	29.7	1.01 (0.79-1.30)	.907	42.8	1.10 (0.89-1.36)	.394
2022	27.5	0.93 (0.72-1.18)	.538	47.7	1.12 (0.91-1.37)	.292
2023	29.2	1.06 (0.83-1.36)	.636	46.2	0.97 (0.79-1.18)	.745
40-44 y						
2017	29.2	—		44.2	—	
2018	31.6	1.08 (0.83-1.41)	.552	46.3	1.05 (0.84-1.31)	.684
2019	31.2	0.99 (0.77-1.27)	.919	45.8	0.99 (0.80-1.23)	.936
2020	26.3	0.84 (0.65-1.09)	.200	34.7	0.76 (0.60-0.95)	.017 ^c
2021	26.9	1.02 (0.78-1.34)	.870	27.4	0.79 (0.61-1.02)	.072
2022	31.0	1.15 (0.90-1.49)	.270	30.5	1.11 (0.86-1.45)	.419
2023	28.7	0.93 (0.72-1.18)	.539	31.5	1.03 (0.81-1.33)	.793
45-49 y						
2017	25.6	—		25.3	—	
2018	28.5	1.11 (0.85-1.46)	.431	33.3	1.31 (1.00-1.73)	.048 ^b
2019	27.3	0.96 (0.74-1.25)	.756	33.7	1.01 (0.79-1.30)	.911
2020	18.3	0.67 (0.49-0.90)	.008 ^c	18.3	0.54 (0.40-0.73)	< .001 ^c
2021	21.5	1.18 (0.86-1.61)	.315	21.9	1.19 (0.86-1.66)	.287
2022	23.6	1.10 (0.82-1.48)	.524	28.1	1.28 (0.96-1.73)	.096
2023	21.3	0.90 (0.67-1.21)	.482	29.1	1.04 (0.79-1.36)	.793
50-54 y						
2017	14.4	—		12.5	—	
2018	14.8	1.03 (0.71-1.50)	.861	13.9	1.11 (0.74-1.66)	.606
2019	17.7	1.19 (0.84-1.70)	.321	18.9	1.36 (0.95-1.96)	.097
2020	12.9	0.73 (0.50-1.04)	.084	15.6	0.83 (0.58-1.17)	.289
2021	14.1	1.10 (0.75-1.61)	.622	13.0	0.84 (0.57-1.22)	.356
2022	15.0	1.06 (0.74-1.53)	.746	14.3	1.10 (0.75-1.62)	.640
2023	16.1	1.07 (0.76-1.52)	.697	19.3	1.35 (0.95-1.93)	.095
55-59 y						
2017	7.9	—		6.7	—	
2018	7.7	0.97 (0.59-1.59)	.890	8.3	1.24 (0.73-2.14)	.422
2019	10.1	1.32 (0.83-2.12)	.242	7.7	0.93 (0.56-1.55)	.780
2020	6.6	0.66 (0.40-1.06)	.090	4.8	0.62 (0.34-1.10)	.110
2021	7.2	1.09 (0.64-1.84)	.759	5.4	1.12 (0.59-2.14)	.724
2022	8.3	1.15 (0.70-1.91)	.577	7.3	1.37 (0.77-2.46)	.291
2023	8.5	1.02 (0.63-1.65)	.947	6.9	0.94 (0.54-1.62)	.828

^aIncidence per 100,000 members.^bPoisson regression model. The prior year was used as the reference year in regression models.^cStatistically significant with a $P < .05$, meaning a significant change in incidence from the prior year.

CI = confidence interval. IRR = incidence rate ratio.

increase in the time to ACLR after injury. Although the median time to surgery increased by 1.6 months in 2020 compared to the previous year, the

following year saw a correction to 3.2 months. This was likely due to a ramping up of ACLR after the suspension of elective procedures, especially in the

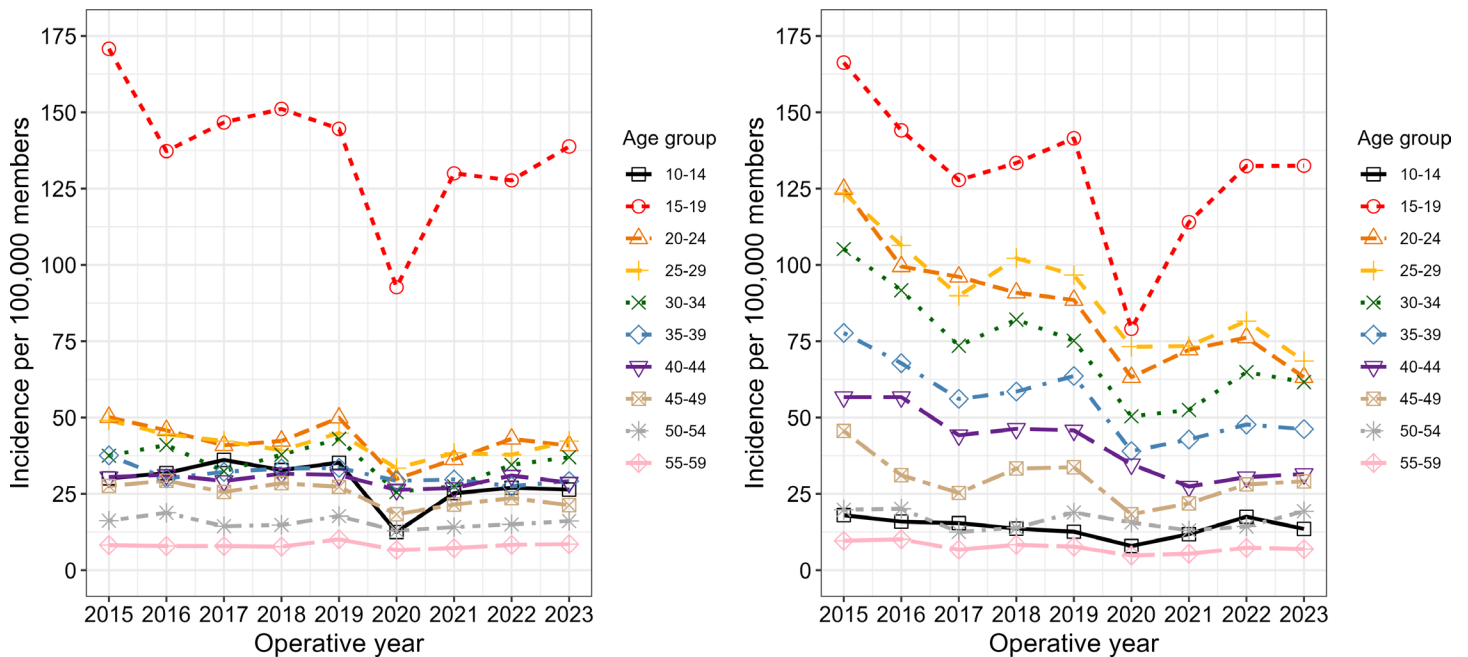


Figure 3: Annual incidence of primary anterior cruciate ligament reconstruction in a US-based health care system, stratified by age (2017–2023). Left panel presents incidence for women and right panel presents incidence for men.

groups ranging from 10 to 19 years old. There was an 81% and 42% increase in incidence from 2020 to 2021 for the 10–14 and 15–19 years of age groups, respectively, far outpacing other age groups.

Interestingly, the present study indicates ACLR incidence is still lagging behind, as ACLRs in most of the age groups have not yet fully returned to levels similar to those prior to the pandemic. Conversely, Kiani et al⁶ reported ACLR incidence had recovered to pre-pandemic levels by 2021. Similarly, a study out of Norway indicated pediatric ACLR incidence had not only recovered to pre-pandemic levels but surpassed them by 2021.³ The discrepancy between these other studies and the present one is likely a result of different data collection practices and national differences in practice pattern. Kooy et al³ suggested different trends are likely to be seen in Norway because conservative management for ACL tears had been the mainstay for treatment until recently. Since the passing of Title IX in the United States in 1972, female participation in organized sports has increased. Mall et al¹⁴ noted a corresponding trend of increasing female ACLR cases in the United States. This trend toward increased ACLR incidence in female patients was also reported in Canada.² The findings in the present study are in line with this increase; women accounted for 42.8% of the ACLR performed in 2017 and 45.3% in 2023. The

shutdown caused by the COVID-19 pandemic had a similar impact on ACLR utilization for both genders; the sharpest decline was seen in the aged 15–19 years group.

Good clinical outcomes, low donor site morbidity, and advantageous structural properties have supported the use of QT autograft over the past decade.^{19–22} Over a similar time frame, HT autograft has lost a proportional level of support with reports of increased failure rates and relatively poor clinical outcomes.^{23,24} Although there has certainly been evidence to the contrary for both QT and HT autograft,^{25–28} the international consensus is best seen in graft usage trends in various registries. In their most recent annual report, New Zealand's ACL registry reported a continued rise in the use of QT tendon and BPTB autografts, with a corresponding decline in the use of HT autografts.²⁹ The Norwegian study also reported a similar trend.³ The present study is in agreement with this international increase in QT and decrease in HT autograft selection.

In previously published work using the same registry as the present study, allografts were the predominant graft of choice for ACLR, accounting for 42.4% of all grafts used.³⁰ Other studies have confirmed that allograft usage in ACLR is plagued by higher graft re-rupture rates and poor clinical

outcomes.^{31–33} Allograft use is less common in countries outside of the United States, accounting for as low as 0.3% of grafts used for ACLR.²⁸ In the United States, a previous report indicated a change in practice patterns, with allografts being reserved for older and less active patients.³⁴ Within the authors' own integrated health care system, allograft utilization decreased, following a coordinated allograft reduction program, to only 2.4% of the ACLRs performed by 2017.³⁵ In the United States, allografts will likely always have a role in ACLR, as they are reserved for a specific patient population and with mounting evidence regarding nonirradiated allografts.³⁶

STRENGTHS AND LIMITATIONS

The data sources are this study's greatest strength. Data were gathered within the Kaiser Permanente health care system, with an over 90% participation rate among surgeons. The data were collected prospectively in a standardized manner with continuous surveillance to ensure quality.

The present study was retrospective and observational in nature, and there are limitations. The first limitation was the authors' inability to determine causality, given the nature of the study. The authors' results only report on associations. In addition, the authors were unable to perform age- and gender-adjusted analyses for trends in hybrid and QT ACLR utilization, given the small number of procedures performed with these grafts annually. However, the crude trends still provide some information on utilization of these grafts, particularly the increasing use of QT ACLR. Second, as the scope of the study was limited to the incidence of ACLR, the authors cannot comment on the incidence of ACL tears. Prior work suggests ACLR incidence may not be a reliable surrogate for actual ACL injuries; however, the primary purpose of the present study was to determine the impact COVID-19 had on practice patterns for ACLR, not necessarily ACL injury rates.³⁷ Third, outcome measures following ACLR, such as reoperation or complications, were not considered. To fully assess the impact of the pandemic on this front, further follow-up is required. Fourth, members of the this study's health care system are largely specific to the western United States; findings of the present study may not be generalized to other areas of the United States where patient demographics and practice patterns, as well as responses to the pandemic, vary.

Conclusion

The shutdown of elective procedures due to the COVID-19 pandemic resulted in a sharp decrease in ACLR utilization within the authors' US-based integrated health care system. The authors' most recent numbers suggest that the United States is still in the recovery period and has not yet returned to the pre-pandemic levels of ACLR utilization observed in 2019. The decline during the pandemic disproportionately diminished procedures performed in pediatric patients, affecting both genders. Graft use patterns did not change due to the pandemic, as pre-pandemic trends continued into the post-pandemic period.

Data-Sharing Statement

Underlying data are not available.

Supplementary Materials

Supplemental material is available at: <https://www.thepermanentejournal.org/doi/suppl/10.7812/TPP/25.061#supplementary-materials>.

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